

Guidance on AI use for laboratory workshop reports in CHEM4032

Approved AI Tool: Students may use **Microsoft Copilot** for some aspects of this assessment, accessed through your University of Nottingham account. This ensures data protection compliance and equitable access for all students. Students may use (with the appropriate caution) bespoke software which includes AI tools for processing molecular data, building machine learning models for predicting various properties and generative AI as it relates to molecular problems. Students may also use machine learning platforms, such as SciKit-Learn and TensorFlow.

Prohibited tools include: ChatGPT, Claude, Gemini, or any other AI platforms not provided by the university.

All reports must be written in English. Translation software must not be used.

Quality and Learning Considerations

Remember that AI tools are not always accurate or reliable, therefore AI output requires critical evaluation:

- Verify accuracy of all AI-suggested information.
- Question assumptions embedded in AI responses.
- Apply module guidelines to evaluate AI suggestions critically.
- Develop your own insights, building on but going beyond AI input.

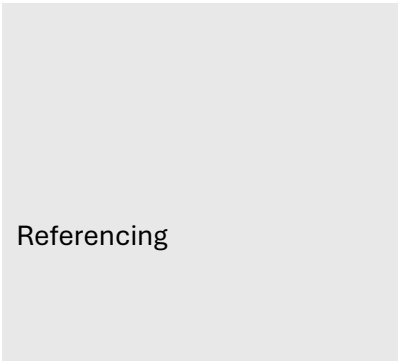
Academic Integrity Expectations

- Original thinking: Your analysis, conclusions, and strategic recommendations must reflect your own critical thinking and application of module concepts.
- Source verification: You must verify any factual information suggested by AI through credible academic or industry sources.
- Intellectual honesty: If AI significantly influenced a particular argument or framework, acknowledge it clearly (following the examples below).
- Attribution: you must properly document AI use (see instructions below).
- **Unacknowledged AI use (including the use of translation software) may constitute false authorship under the university's Academic Misconduct policy.**

AI may enhance your learning process, not replace critical thinking. The strongest reports will demonstrate how you have used AI as a starting point for deeper analysis rather than accepting AI output uncritically.

As the human in the loop, you remain responsible for all content submitted, including anything generated or assisted by AI.

Key academic activities for this assessment.	AI use is Essential/ Optional/ Prohibited for this task	Guidance on AI tools and what to use them for.	What information and/or evidence you should keep from this part of the process.	Where you can learn to do this, with or without AI as appropriate.
Finding and selecting appropriate sources of information and knowledge.	Optional	You may use Microsoft Copilot to help you, however you must critically review the use and outputs.	If you choose to use AI in this way, you should keep a full log of your dialogue and submit this in an appendix. All references must be to the research literature article, rather than an AI prompt.	If you include references in your report, you are responsible for checking that the citations are correct, and that any information taken from them is properly attributed.
Developing a plan or outline or the report.	Prohibited			Please use the report template on Moodle.
Creating visuals to communicate ideas.	Prohibited			You should use chemistry drawing software, e.g. ChemDraw or ChemSketch, to produce chemical structures. You may use any other tool (Excel, Matlab, Python, Prism etc.) to generate scientific images, but these should not be AI generated.
Writing a draft.	Prohibited			
Performing calculations and analysing data.	Optional	You may use bespoke software which includes AI tools for processing molecular data, building machine learning models for predicting various properties and generative AI as it relates to molecular problems. Students may also use machine		More details are included in the instructions for each of the laboratory workshops.



Optional

learning platforms, such as SciKit-Learn and TensorFlow. You may use Microsoft Copilot to help you; however, you must critically review the use and outputs.

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AI tools can be used to format reference lists. Software for referencing such as EndNote or Mendeley can also be used.